

Morphology, Syntax, and Semantics of Mongolian Causatives: An Interface Analysis¹

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Abstract: While causatives have been one of core issues in Mongolian linguistics, many aspects of them have been understudied. One important reason for this lies in the fact that typological approaches and interface analyses have been rare. The main purpose of this paper is to classify and characterize Mongolian causatives according to their morphological, syntactic and semantic properties. Adopting a typological method, this paper demonstrates that to characterize a given verb as a causative, its semantics is the key factor and causative semantics is syntactically reflected; while morphology also reflects causative semantics, it is not a crucial factor for characterizing causatives in Mongolian.

Keywords: Mongolian, causative, morphology, syntax, semantics

1. Introduction

Most, if not all, of previous studies including Poppe (1954: 170-171), Sanzheyev (1973: 85), Nasanbayar (1982: 351-352), Daobu (1983: 45-46), Cinggeltei (1999: 285-286), Kullman and Tserenpil (2005: 117-123) and Janhunen (2012: 249-250) have identified a given verb affixed by any of *-uul*, *-lg* and *-aa* as “үйлдүүлэх үйл”² (meaning “causative verb”), regardless of whether their semantic and syntactic properties qualify them for being causative or not. In this paper, I label a verb that is morphologically causative, with or without causative semantics, “morph-causative”. I also use “sem-causative” to refer to a verb that has causative semantics but may or may not have causative morphology.

(1) Morph- vs. Sem-causative

Classification	Characterization	Example
morph-causative	have causative morphology; may or may not have causative semantics	<i>bai-gul</i> ‘build’ <i>orč-uul</i> ‘translate’
sem-causative	have causative semantics; may or may not have causative morphology	<i>bai-lg</i> ‘let be’ <i>šat-aa</i> ‘burn (tr)’

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² Allomorphs of the causative suffixes include *-uul/-gul*, *-lg*, *-ga/-aa* and their allophonic variants. See the appendix for details.

A fact that has been ignored in the literature is that not all morph-causatives in Mongolian are semantically causative. Importantly, genuine causatives are sem-causatives rather than morph-causatives in a typological context. That is, semantics and syntax, rather than morphology, play a key role in deciding whether a given verb is causative or not. Accordingly, those that have morphologically analytic structure, i.e. V plus CS (V-cs),³ but fail to satisfy the semantic criterion of causation are “fake causatives” and those that do not fail are genuine and/or quasi causatives. Throughout the remaining of the paper, the term “causative” is used to refer to genuine or quasi causatives unless specified.

(2) Fake vs. Genuine vs. Quasi causative

Classification	Characterization	Example
fake	morph-causative without causative semantics	<i>orč-uul</i> ‘translate’ <i>xania-lg</i> ‘cough’ <i>aıld-gaa</i> ‘report to a superior’
genuine	sem-causative with conventional causative morphology (-uul, -lg, -aa)	<i>bari-uul</i> ‘cause to arrest’ <i>inee-lg</i> ‘cause to laugh’ <i>šat-aa</i> ‘burn (tr)’
quasi	sem-causative with the unconventional morphology (-l) or without causative morphology	<i>tas-l</i> ‘break of (tr)’ <i>nee</i> ‘open (tr)’

Criteria of causation denoted by a verb are essentially semantic. As discussed by the influential studies on characterizing causatives such as Comrie (1976), Shibatani (1976), Frawley (1992) and Song (1996), the event denoted by a given causative predicate must involve two sub-events, one being a causing event and another being a caused one. The causing event leads to introduction of a causer argument, which refers to someone or something that initiates or controls the activity (Dixon 2000: 31). One of the syntactic (or semantic-syntactic) properties of causatives is installment of a new argument onto the corresponding noncausative verb, as discussed by Comrie (1976). Here, the notion of “corresponding” is non-ignorable. Although, Comrie (1976) does not explicate what he means exactly by “corresponding”, it is implied in his discussion and examples that X is said to be corresponding with Y if one of them involves the same valency and the same semantics as the other. That is, if X is V and Y is V-cs, then the valency and semantics of V does not vary between X and Y.

Given this, morph-causatives like *bai-gul* ‘establish, build’ are not causative per se since they do not denote events that can be interpreted as involving two sub-events; that is, they do not mean “cause to do”. *Bai-gul*, for example, cannot be paraphrased by an expression meaning “X cause Y to exist/be”. This becomes clearer when we compare *bai-gul* with *bai-lg*, which share the same root *bai* ‘be, exist’. *Bai-lg* is clearly causative, meaning “let be”, and therefore is both morph- and sem-causative. As indicated by the English translations, only in *bai-lg*, the

³ “CS”, short for “causative morpheme”, is used as cover label for *-uul/-gul*, *-lg*, *-ga/-aa* (including their allophonic variants) and “V-cs” for any verb with CS, whether it qualifies as causative or not.

meaning of *bai*, i.e. “be, exist” remains the same. By contrast, semantic drift has taken place with *bai-gul*.

On the other hand, not all verbs that would not be treated as causative (due to their lack of *-uul*, *-lg* or *-aa*) are sem-causatives and therefore may be identified as causative verbs. For example, *tas-l* ‘break off (transitive)’ can be interpreted as “to cause to break off (intransitive)”. Sem-causatives are subject to distinctions made in four different ways. The first distinction is made between marked and unmarked causatives. By “marked” I mean that a given sem-causative has morphologically analytic structure of R (V or others) plus CS and by “unmarked” I mean that a given sem-causative has no morphological indication as such. Sem-causatives can also be divided into syntactic and lexical causatives. Dative and instrumental causatives have the causee argument marked as dative and instrumental respectively. Syntactic causatives are those that satisfy the criteria given in section 3.4 and lexical causatives are those that do not. The last distinction is made between deverbal and non-deverbal causatives, the former being formed on a verbal base and the latter on a non-verb root.

(3) Marked vs. Unmarked causative

Classification	Characterization	Example
marked	sem-causative with causative morphology	<i>bari-uul</i> ‘cause to arrest’ <i>inee-lg</i> ‘cause to laugh’ <i>šat-aa</i> ‘burn (tr)’ <i>tas-l</i> ‘break of (tr)’
unmarked	sem-causative without causative morphology	<i>nee</i> ‘open (tr)’

(4) Dative vs. Instrumental causative

Classification	Characterization	Example
dative	sem-causative with dative causee	<i>Bat-d süü uu-lg</i> ‘give Bat some milk to drink’
instrumental	sem-causative with instrumental causee	<i>Bat-aar süü uu-lg</i> ‘let Bat drink milk’

(5) Syntactic vs. Lexical causative

Classification	Characterization	Example
syntactic	See section 3.4.	<i>xel-üül</i> ‘ask to say’ <i>xii-lg</i> ‘cause to do’
lexical	See section 3.4.	<i>erg-üül</i> ‘turn (tr)’ <i>cai-lg</i> ‘whiten (tr)’ <i>šat-aa</i> ‘burn (tr)’ <i>tas-l</i> ‘break of (tr)’

(6) Deverbal vs. Non-deverbal causative

Classification	Characterization	Example
deverbal	formed on a verbal base, transitive or intransitive	<i>šat-aa</i> ‘burn (tr)’ <i>xii-lg</i> ‘cause to do’ <i>xög-ž-üül</i> ‘develop (tr)’ <i>bag-s-gaa</i> ‘reduce (tr)’
non-deverbal	derived out of a non-verb root	<i>tas-l</i> ‘break of (tr)’

Discussion throughout the remaining of the paper covers all these types of causative. Section 2 focuses on the morphological aspects of morph- and sem-causatives as well as of the causative-anticausative alternation. Section 3 discusses the syntactic characteristics of causative and fake causatives, focusing particularly on the interactions of thematic and conceptual roles of arguments, their case and their grammatical functions. Section 4 is devoted to working out a basic characterization of causatives, in accordance with Dixon’s (2000) parameters. Section 5 concludes this paper.

2. Morphology

2.1. Fake causative: morpho- but not sem-causative

As already mentioned in the introduction, fake causatives are those that have morphologically analytic structure, i.e. V plus CS, but fail to satisfy the semantic criterion of causation. Fake causatives are simply non-causative transitives. Some examples are listed below.

(7) Fake causatives:⁴

	“Hypothesized causative” meaning	True meaning
<i>bai-gul</i>	cause to be	build, establish
<i>orč-uul</i>	cause to revolve	translate
<i>zeel-üül</i>	cause to borrow	lend
<i>suu-lg</i>	cause to sit	suffer from diarrhea
<i>aıld-gaa</i>	cause to waffle	report to a superior

As indicated by the English translations, the verbs listed are not causative verbs. A salient characteristic of them is that semantic drift takes place between the base verb and the formed one. For example, while *orč-uul* means “translate”, its base verb *orč* means “revolve”. Notably, *orč-uul* fails to mean “cause to revolve”. Their lack of causative semantics can be syntactically tested. The base verb resists adverbial modification. For example, *saixan* in *saixan orč-uul* ‘translate carefully’ modifies the whole event of translating. It does not make sense to say that it modifies a sub-event like, for example, “revolving a word or expression from one language

⁴ Some of the verbs listed such as *zeel-üül* ‘lend’, and *suu-lg* ‘cause to sit’ can also be causative verbs. See section 3.2 for detailed discussion of this issue.

to another”. The failure of the test indicates that the event does not involve a caused event and therefore a causing event is not available either.

2.2. Marked causative: morpho- and sem-causative

Marked causatives are those that are sem-causatives with a morphologically analytic structure of V (N, Adj, or Adv) plus CS. Marked causatives are divided into four subtypes including V-*uul*, V-*lg*, V-*aa* and R-*l*, R being Adv, Adj or V.

While V-*uul*, V-*lg* and V-*aa* types of causative are formed on the basis of a verb, R-*l* causatives are created by affixing *-l* to what can be identified as an adverbial particle, e.g. *tas* and *xag*, which are labeled “state-denoting adverbs” in traditional Mongolian grammar, to an adjective or to a verb.

Traditional grammarians have never treated *-l* as a causative marker, however. But the semantic and syntactic properties of R-*l* point to the fact that it is a causative. Firstly, R-*l* denotes an activity with a result state, which can be analyzed as “causation of inchoative action”. An activity with a result state instantiates a sub-event. Because such an activity is not spontaneous in the case of R-*l*, it is natural to claim that there is an unspecified event that serves as a external force to make the activity take place. Secondly, R-*l* participates in the causative-anticausative alternation, with the anticausative being R-*r*. R-*l* takes two arguments and R-*r* takes one.⁵

(8) Adv-*l*:

<i>tas-l</i>	‘cause to break off’	<i>bialb-l</i>	‘smash’
<i>xag-l</i>	‘cause to break’	<i>niac-l</i>	‘smash’
<i>zad-l</i>	‘cause to open, unpack’	<i>sox-l</i>	‘make blind’
<i>coo-l</i>	‘make a hole on’	<i>cöm-l</i>	‘cause to cave in’
<i>xug-l</i>	‘cause to break off’	<i>nebt-l</i>	‘penetrate’
<i>balb-l</i>	‘smash’		

(9) Adj-*l*:⁶

<i>tod-l</i>	‘clarify’
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(10) V-*l*:

<i>ebd-l</i>	‘damage’
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Among the three subtypes, Adv-*l* seems to be the prototypical one. Adv-*l* differs from Adv-*l* and V-*l* in that Adv in Adv-*l* co-occurs with the function verb *tabi*, which may well be characterized as a light verb, e.g. *xag tabi* ‘break off (tr)’, while Adj-*l* and V-*l* resist *tabi*. That is, importantly, Adv-*l* and Adv-*tabi* are used interchangeably. Interestingly, the anticausative –

⁵ Verbs in the form of V-*l/r* are called “medial verbs” in Poppe (1951: 51).

⁶ *Tod* can also be Adv since it can co-occur with the light verbs *tabi* and *oč*. Moreover, it can be duplicated and modify a verb as those in (8) do. *Tod* is labeled Adj here since it can be used as a noun modifier.

r, too, is interchangeable with a light verb, namely, *oč*, e.g. *xag oč* ‘break off (intr)’. This is not true for Adj-*r* and V-*r*.

2.3. Unmarked causative: not morpho- but sem-causative

What can be identified as unmarked causatives are verbs such *nee* ‘to open’, *xaa* ‘to close’ and *bar* ‘to use up’. They are unmarked since they contain no suffix and are equivalent morphologically to their inchoative alternants.

(11) Unmarked causatives:

<i>nee</i>	‘open (tr)’
<i>xaa</i>	‘close (tr)’
<i>bar</i>	‘use up, cause to exhaust’

Unlike the other types of causative, unmarked causatives allow passivization with an active meaning.

(12) Fake passives:

<i>nee-gd</i>	‘open (intr)’
<i>xaa-gd</i>	‘close (intr)’
<i>bar-gd</i>	‘run out’

In that case, the passive suffix PS functions as an inchoative marker and affixing PS to V does not instantiate genuine passivization. In my own work (Bai 2023), I explicated that PS has undergone or is undergoing a process of losing its passivizing function, thereby to produce a “fake passive”. PS as a fake passive marker, however, is not possible with the aspectual suffix *-ž bai*, which can select for the anticausative *nee* and *xaa* to denote a resultative state and the causative *nee* and *xaa* to denote an ongoing action.⁷

- (13) a. Üüd *nee-ž bai-na*.
 door-NOM⁸ open-Ž BAI-NPST
 ‘The door is open.’
 b. Üüd *xaa-ž bai-na*.
 door-NOM close-Ž BAI-NPST
 ‘The door is closed.’

2.4. Types of causative-anticausative alternation

⁷ The acceptability of these alternants varies according to dialects.

⁸ The abbreviations used for glossing in this paper include ABL: ablative, ACC: accusative, CS: causative, CVB: converb, DAT: dative, INS: instrumental, NPST: non-past, PS: passive, PST: past, RX: reflexive-possessive, and VOL: volutative.

Hapelmath (1993) divided causative-anticausative alternations observed cross-linguistically into five types according to their morphological structure.

(14) Five types of causative alternation (adapted from Haspelmath 1993: 91-92):

a. Anticausative-based alternation: ($V_{\text{inch}} + \text{AF} \rightarrow V_{\text{caus}}$)

Arabic: *darasa* 'learn'
darrasa 'teach'

b. Causative-based alternation ($V_{\text{caus}} + \text{AF} \rightarrow V_{\text{inch}}$)

Russian: *katat'* 'roll (tr)'
katat'-sja 'roll (intr)'

c. Equipollent alternation: ($V + \text{AF1} \rightarrow V_{\text{caus}}$; $V + \text{AF2} \rightarrow V_{\text{inch}}$)

Japanese: *atum-aru* 'gather (intr)'
atum-eru 'gather (tr)'

d. Suppletive alternation: ($V1_{\text{caus}}$; $V2_{\text{inch}}$)

Russian: *goret'* 'burn (intr)'
žeč 'burn (tr)'

e. Labile alternation: ($V_{\text{inch/caus}}$)

Greek: *svíno* 'go out'
'extinguish'

Viewed from Haspelmath's (1993) classification, causative-alternations in Mongolian fall into three types. Anticausative-based alternation features formation of the causative alternants from the (anti)causative one by employing one of the causative suffixes *-uul*, *-lg* and *-aa* (including their allophones). Equipollent alternation includes those that are formed from the adverbial particles by adding *-l* and *-r*. Labile alternation includes those without any morphemes indicating the causative alternation.

(15) Three types of causative alternation in Mongolian:

a. Anticausative-based alternation:

šat 'burn (intr)'
šat-aa 'burn (tr)'

b. Equipollent-based alternation:

xag-r 'break (intr)'
xag-l 'break (tr)'

c. Labile alternation:

nee 'open (intr)'
'open (tr)'

3. Syntax

3.1. Argument role and causative-anticausative alternations

One syntactic diagnostics of causatives is their ability to participate in a causative-anticausative alternation and a defining property of causative-anticausative alternations is role-installing. That is, a new argument, causer, is introduced in the derivation of a causative (or anticausative) out of its anticausative (or causative) alternant, with the role of the original argument(s) remaining unchanged with respect to the base alternant.

Three patterns of causative-anticausative alternation can be attested in Mongolian. All they feature derivation of the causative alternant out of the anticausative one. In the first two patterns, the anticausative alternant is intransitive, with the sole argument being either a patient or an agent. In the third pattern, the anticausative alternant is transitive, out of which the causative is derived.

(16) Deriving causative out of inchoative; role of Y remaining [P (patient)]:

<i>šat</i>	‘burn (intr)’	< Y _[P] >
<i>šat-aa</i>	‘burn (tr)’	< X _[C] , Y _[P] >

(17) Deriving causative out of unergative; role of Y remaining [A (agent)]:

<i>inee</i>	‘laugh’	< Y _[A] >
<i>inee-lg</i>	‘cause to laugh’	< X _[C] , Y _[A] >

(18) Deriving causative out of transitive; role of Z/Y remaining [P]/[A]:

<i>unš</i>	‘read’	< Y _[A] , Z _[P] >
<i>unš-uul</i>	‘cause to read’	< X _[C] , Y _[A] , Z _[P] >

If Y fails to retain the same role, whether V retains the same semantics or not, V-cs is a fake causative and fails to constitute a causative-anticausative pair with V. This happens in the case that V in V-cs is a transitive. I exemplify this with *zeel-үүл* ‘lend’, which is derived out of *zeel* ‘borrow’. What happens in the derivation of *zeel-үүл* is that what comes as the most prominent argument (i.e. the agent) that is realized as the left-most function (i.e. the subject) switches from Y (the goal/recipient, i.e. the borrower) to X (the source, i.e. the lender). The event itself – Z (the undergoer, i.e. what is borrowed/lent) moving from X to Y – does not change, but the viewpoint of the speaker changes. This is the conceptual semantics of *zeel-үүл* and *zeel*. In a conceptual structure, roles remain all the same. Importantly, however, in the theta semantics, especially when viewed from the perspective of the current argument/theta-role theory, Y undergoes a change of role from [A] to [R] and when this happens *zeel* and *zeel-үүл* do not constitute the alternating pair of causative and anticausative.

(19) Deriving fake causative out of transitive; role of Y changed from [A] to [R (recipient)]:

<i>zeel</i>	‘borrow’	< Y _[A] , Z _[P] >
<i>zeel-үүл</i>	‘lend’	< X _[C] , Y _[R] , Z _[P] >

Zeel and *zeel-üül* may be treated as antonyms in this case, meaning “borrow” and “lend”. Cross-linguistically, *borrow*-verbs and *lend*-verbs do not alternate with respect to (anti)causative; they make up a dative-transitive pair instead. This allows us to treat *zeel-üül* meaning “lend” as a dative verb.

The causative-anticausative alternation, however, obtains for *zeel* and *zeel-üül* in the case that the role of Y does not change in the course of formation of *zeel-üül*, as outlined below.

(20) Deriving causative out of transitive; role of Z/Y remaining [P]/[A]:

<i>zeel</i>	‘borrow’	< Y _[A] , Z _[P] >
<i>zeel-üül</i>	‘cause to borrow’	< X _[C] , Y _[A] , Z _[P] >

Zeel-üül means “cause to borrow” in (21b) and “lend” in (21c). Only that in (21b) makes up a causative-anticausative pair with *zeel*, which means “borrow”.

- (21) a. Badma nom *zeel*-v.
 Badma-NOM book-ACC borrow-PST
 ‘Badma borrowed a book.’
- b. Baatar Badma-gaar nom *zeel-üül*-v.
 Baatar-NOM Badma-INS book-ACC borrow-CS-PST
 ‘Baatar asked Badma to borrow a book.’
- c. Baatar Badma-d nom *zeel-üül*-v.
 Baatar-NOM Badma-DAT book-ACC borrow-CS-PST
 ‘Baatar lent a book to Badma.’

The variation between the *lend* meaning and *cause-to-borrow* meaning of *zeel-üül* is closely related with the choice of case between DAT and INS for Y. We return to this issue in section 3.2.

What we discussed above is a role-demoting case, which is observed with deriving a fake causative out of a transitive. By contrast, role-deleting takes place when a fake causative is derived out of an intransitive. Role-deleting with intransitives is more tricky than role-demoting in that Y fails to retain the same role with respect to the same base verb in the course of formation. In this case, too, causative-anticausative alternation does not obtain, leading to the fact that V-CS is a fake causative. One of typical examples is *orč-uul* ‘translate’.

(22) Deriving fake causative out of intransitive; role of Y remaining [P] with different bases:

<i>orč</i>	‘revolve’	< Y _[P] >
<i>orč-uul</i>	‘translate’	< X _[C/A] , Y _[P] >

To explicate, Y fails to retain the same role [P] with respect to the same base verb *orč*. Once *orč* comes to be affixed by CS, the original argument (call it Y1) of V disappears because the revolving event – what would otherwise be the caused event – is eliminated entirely in *orč-uul*.

That is, the event denoted by V is now not a revolving event and therefore the participant of that event (the argument of V) is not Y1, but rather a different one (call it Y2). So, to say that Y1 is not present in *orč-uul* is to say that the role of Y1 is deleted. Therefore, Y now represents a participant of the event denoted by V-cs, not of the one denoted by V. This leads to semantic drift that is brought about by “fading out of the semantics of V and fading in of that of V-cs”. Because the “would-be caused event” is eliminated, causation does not obtain. *Orč-uul* cannot be interpreted as “cause to V”, where V represents the “would-be caused event”. The only possible event *orč-uul* can denote is one of translating, which is mono-eventive per se. It then follows that V and V-cs do not make up a causative-anticausative pair in this case.

One might prefer a different explanation described as follows. The argument Y is not eliminated because what serves to be part of the semantics of *orč-uul* ‘translate’ is a frame of event rather than an event itself. For *orč-uul* here, “frame of event” can be understood as a frame that has an argument slot for Y that undergoes a change of location, thereby shaping a “metaphorical event” that could be described as “revolving”. This gives us an instruction such as “To translate X is to cause X revolve from one language to another”. Under this analysis, the role of Y would remain the same, that is, [P]. One limitation of this analysis is that it fails to account for what renders semantic drift between *orč* ‘revolve’ and *orč-uul* ‘translate’. Moreover, this analysis would render a bi-eventive structure for *orč-uul*. Translating, however, is a simple event with a mono-eventive structure. *Orč-uul* fails the adverbial-modification test. Nor is it available for an interpretation such as “cause X to revolve from one language to another” or a similar one.

Still another explanation may be competitive. What happens in deriving V-cs out of V is deletion of the actor role [AT] on Y, Y itself remaining the same. [AT] is deleted on Y and installed on X. The role-deleting that takes place with Y leads to a semantic drift. This analysis, again, is in favor of the role-deleting approach to the relevant semantic drift.

All in all, pairs like *orč-uul* and *orč* do not make up a causative-anticausative alternation due to semantic drift brought about by role-deleting on Y.

To sum up this section, role-installing (valency-increasing), role-deleting (valency-reducing), role-demoting and role-merging can be observed with formation of V-cs, which may either be a causative or not. When V-cs is a causative, only role-installing takes place; when it is not a causative, role-installing, role-deleting and role-merging take place in the case of deriving V-cs out of intransitives. In the case of deriving V-cs out of transitives, role-installing and role-demoting take place. From this it follows that role-installing is not a sufficient condition but a necessary one for deriving causatives and that ability to participate in a causative-anticausative alternation is suggested to be added as another necessary condition. Needless to say, semantically, causatives involve two sub-events.

3.2. Case and distinction of instrumental vs. dative vs. fake causatives

Case (partially) mirrors the role of an argument and the event semantics related to it. For the first example, when *zeel-üül* is used as a causative verb, it requires Y be assigned INS and

when it is not, Y is necessarily assigned DAT. In the latter case, *zeel-üül* does not have causative semantics but dative semantics, the sentence instantiating a dative construction.

(23) Dative vs. Causative verbs

	Meaning	Voice type	Valence type	Case variation
<i>zeel</i>	‘borrow’	transitive	< Y _[A] , Z _[P] >	< NOM, ACC >
	↓			
<i>zeel-üül</i>	‘lend’	dative	< X _[C] , Y _[R] , Z _[P] >	< NOM, DAT, ACC >
	‘cause to borrow’	causative	< X _[C] , Y _[A] , Z _[P] >	< NOM, INS, ACC >

Obviously, the same form participates in two distinct pairs, dative-transitive and causative-transitive (i.e. causative-anticausative). I label such verbs “bi-varient verbs” in this paper. Bi-varient verbs also include *težee-lg*, *üz-üül* and *med-üül* among others.

Not all V-cs are bi-variants, however. Sometime V-cs can only be causative, lacking a dative variant (24), or can only be dative, lacking a causative variant (25).

(24) X(NOM), Y(INS/*DAT), Z(ACC):

<i>xel-üül</i>	‘X cause Y to say Z’
<i>xii-lg</i>	‘X cause Y to do Z’
<i>zas-uul</i>	‘Z cause Y to fix Z’

(25) X(NOM), Y(*INS/DAT), Z(ACC):

<i>med-üül</i>	‘X report Z to Y’
<i>xürt-ee</i>	‘X give Z to Y’
<i>ol-g</i>	‘Z grant Z to Y’

As exemplified in (25), the dative argument of the verbs is interpreted as a recipient or beneficiary of V-cs, although it may be inferred that that argument is the original agent-like argument of V. Note that for V-cs the dative argument is not an agent in (25), unlike the case of the instrumental argument in (24). In (24), the instrumental argument is an argument of V since it is assigned a role by V; it is not an argument of V-cs, the verb complex made up from V and the suffix. By contrast, in (25) the dative argument, strictly speaking, is not an argument of V any longer since it acquires a new role from V-cs. The suffix CS has nothing to do with the role of the argument.

Importantly, DAT-taking V-cs can have causative and dative properties for the same predicate, i.e. *uu-lg*.

- (26) Yaaxdan či minii xöbүүн-d doloon ayag airag uu-lg-v?
 why you my son-DAT seven cup koumiss-ACC drink-CS
 ‘Why did you give my son seven cups of koumiss to drink?’

In (26) the speaker's son who performs the drinking action is also a receiver of the milk which was "given" to him by the addressee. The appropriate interpretation of such V-cs is "X gives Z to Y and Y acts on Z", which may correspond to the English "give to do" construction. The role played by Y is ambiguous between [A] and [R]. If we, however, are adherent to Chomsky's (1981) Thematic Criterion, only one of the two roles is possible. Which one to decide on depends on which one is more salient than the other. However, the salience of the roles is not self-evident. If we ignore Thematic Criterion, then [A] and [R] are both active on *minii xöbүүн* in (26). This leads us to treat *uu-lg* as a dative-causative verb. The two sub-events required by causality can be tested by scope-taking adverbials like *xurдан* 'quickly'. This adverb is ambiguous between modifying the causing event and modifying the caused event in (27).

- (27) Bi Badma-d em-g-ni xurдан uu-lg-v.
 I Badma-DAT medicine-ACC-RX quickly drink-CS-PST
 'I quickly gave the medicine to Badma to drink.'
 'I gave the medicine to Badma to quickly drink.'

Such ambiguity is not available for those in (24) and (25), which leads to the difference between dative-causatives and mono-variant verbs including dative V-cs and causative V-cs. It is noted that dative-causative verbs are also different from bi-variant V-cs, e.g. *zeel-үүл*.

Now we have three roles associated with DAT in V-cs including [R], [F]⁹ and [A], which interact with other factors to yield five types of (sub)construction.

(28) Five types of V-cs with DAT

	Member	Thematic role of Y	Conceptual role of Y	Grammatical function of Y	Voice meaning
Type 1	<i>med-үүл</i>	R (recipient)	unspecified	indirect object	active
Type 2	<i>šat-aa</i>	F (force)	cause	adjunct	active
Type 3	<i>uu-lg</i>	A (agent)	causee	embedded subject	active
Type 4	<i>id-үүл</i>	A (agent)	unspecified	adjunct	quasi passive
Type 5	<i>zagn-uul</i>	A (agent)	unspecified	adjunct	passive

- (29) Bi ene yavdal-ig darag-d-aa med-үүл-v. (Y as [R])
 I this matter-ACC leader-DAT-RX know-CS-PST
 'I reported this matter to my superior.'

- (30) Bi xuučin nom-uud-oo gal-d šat-aa-v. (Y as cause [F])
 I old books-ACC-RX fire-DAT burn-CS-PST

⁹ I tentatively take the role of Y in this case as [F] (for "force"), intending that a given event takes place by force of something. Importantly, that thing inducing the event itself is natural force per se. Though such force seems to have not been formulated as a thematic role, the semantic and syntactic behaviors of such V-cs in Mongolian suggest that force is a thematic role. It is noted that the role of Y (i.e. force [F] here) is distinct from those roles that were identified in the standard theta-role theory.

‘I burned my old books in the fire.’

(31) Bi Badma-d süü uu-lg-v. (Y as causee [A])

I Badma-DAT milk drink-CS-PST

‘I gave milk to Badma to drink.’

(32) Bi möng-öö xulgai-d av-uul-v. (X as [A])

I money-ACC-RX thief-DAT steal-CS-PST

‘I got my money stolen by a thief.’

(33) Bi bagš-d-aa zagn-uul-v. (X as [A])

I teacher-DAT-RX scold-CS-PST

‘I got myself criticized by my teacher.’

In type 1, as discussed above, V-cs has dative semantics, rendering Y being assigned [R]. Therefore, the conceptual (not thematic) role it has is unspecific.¹⁰ That is, Y is neither causer nor causee. Y is realized as an indirect object. In type 2, Y, assigned [F], is a cause, rather than a causer or a causee. This type is the only one that sometimes allows V, the anticausative variant, to take the dative element, Y. This means that Y is not relevant to CS. That is, the role of Y does not undergo change and does not lead to the causative semantics of V-cs, either. Y is realized as an adjunct, not as an indirect object. In type 3, Y is assigned [A] by virtue of being the performer of the action, for example, of drinking the medicine. As we have seen, however, the role of Y can also be construed as [R]. It might be likely that Y is assigned a complex role [A+R]. For the moment, I take it to be [A]. In type 4, Y is assigned [A] by virtue of being the performer of the action, for example, of arresting the thief. Y is not a causee since the causee is open for Z, the patient, which realizes as the object. Unlike types 1-3, this type has a passive nuance for Z, that is, “X gets Z done by Y”. The dative Y is an adjunct much like the agentive *by*-phrase in English. No change takes place with the role of Y but a change takes place with its case. Its role is [A] all the time while case is demoted from NOM to DAT. This instantiates the so-called “agent-suppression”. In this sense, suppression is not a matter of theta role but of case. That is, an argument is suppressed if its case is demoted without changing its role. Type 5 differs from type 4 as regards referentiality of Y. In type 4, Y refers to someone or something other than X, while in type 5, Y is coreferential with X and thereby being omitted. A passive reading that obtains in this case is “X gets (oneself) done by Y”. Importantly, the passive alternant of type 5 is more natural than that of type 4.

(34) Bi möng-öö xulgai-d av-t-v. (X as [A])

I money-ACC-RX thief-DAT steal-PS-PST

‘I was stolen my money by a thief.’

(35) Bi bagš-d-aa zagn-gd-v. (X as [A])

¹⁰ What I call “conceptual roles” in this paper refer to causer, causee and cause.

I teacher-DAT-RX scold-PS-PST
'I was scolded by my teacher.'

The difference in nuance between the causative and passive is that in the causative the subject X is responsible for Z, coreferential or not, being done by Y, which is not true in the passive. A slight adversive reading is optionally available for the causative but not for the passive.

The difference between type 5 and type 4 also lies in whether they are interchangeable with a dative construction. For type 4, the the same V-cs may be used as a dative causative predicate, which is not the case for type 5.¹¹ Interestingly, in both type 5 and type 4, the case of Y is perfectly interchangeable with INS, which does not hold in the other types. When Y is assigned INS, the passive meaning is gone.

Interchangeability is also observed between DAT and ACC on the same argument Y. A typical example is *sur-gaa*, which may either be used as a causative verb meaning “cause to learn” or as a dative (non-causative) verb meaning “teach”.

(34) Baatar xüü-d-ee erdem sur-gaa-v.
Baatar-NOM son-DAT-RX skill-ACC learn-CS-PST
'Baatar taught a skill to his son.'

(35) Baatar xüü-g-ee erdem sur-gaa-v.
Baatar-NOM son-ACC-RX skill-ACC learn-CS-PST
'Baatar made his son learn a skill.'

Notably, when meaning “teach”, *sur-gaa* is a fake causative since teaching does not involve causation. Causative semantics is assigned when the case of Y is ACC but not when it is DAT. This becomes explicit when we take *zaa* into consideration. *Zaa*, which also means “teach” but not “cause to learn”, is interchangeable with the dative *sur-gaa*, without a salient difference in meaning, but not with the other.

(36) Baatar xüü-d-ee erdem zaa-v.
Baatar-NOM son-DAT-RX skill-ACC teach-PST
'Baatar taught a skill to his son.'

(37) *Baatar xüü-g-ee erdem zaa-v. (hypothesized structure)
Baatar-NOM son-ACC-RX skill-ACC teach-PST
'Int. Baatar taught his son a skill.'

Semantically, when Y is assigned ACC, X, the subject has a highly manipulative strength with respect to Y and X's “energy” runs directly to Y.¹² This semantics is not true for the DAT-

¹¹ See Janhunen (2012: 250) for relevant discussion.

¹² See Hashimoto (2019) for detailed discussion on the manipulative strength and energy of the subject in

taking *sur-gaa* and *zaa*.

All this means that when Y is assigned ACC, *sur-gaa* qualifies as the causative alternant of *sur*.^{13,14}

3.3. Grammatical function and distinction of argument vs. adjunct

When CS is attached to V to form a causative verb, the original subject S of V becomes an object O of V-cs, being assigned ACC, if V is an intransitive.

(38) $S \rightarrow O$: *šat* (S_i) $\rightarrow$ *šat-aa* (S_j , O_i)

When, however, V is a transitive, the original S of V becomes an embedded subject if INS is assigned to it.

(39) $S \rightarrow ES$: *xii* (S_i , O_j) $\rightarrow$ *xii-lg* (S_k , ES_i , O_j)

S of V becomes an indirect object when DAT is assigned, but which is limited to the case in which Y is not an agent but a recipient.

(40) $S \rightarrow IO$: *zeel* (S_i , O_j) $\rightarrow$ *zeel-üül* (S_k , IO_i , O_j)

When DAT is assigned to Y that is interpreted like the *by*-phrase in English, Y is an adjunct.

(41) $S \rightarrow AD$ (adjunct): *bari* (S_i , O_j) $\rightarrow$ *bari-uul* (S_k , AD_i , O_j)

Linking a particular grammatical function with a particular argument is not as significant as identifying the argument as the bearer of a particular semantic role in that grammatical function is less primitive compared with semantic roles. Specification of a grammatical function other than the subject and direct object is arbitrary to some extent and varies according to analysts or grammar theories. This is also indicated by the difficulties of the current syntactic theories in differentiating adjuncts from arguments, which map into with grammatical functions. The traditional grammarians of Mongolian treat non-NOM and non-ACC elements as indirect objects while theorists are more rigorous in dealing with them. For example, Comrie (1976: 275) suggests that DAT elements in Mongolian are not necessarily indirect objects.

casuatives.

¹³ Alternatively, if indirect or facilitating causation counts as a subtype of causation, DAT in DAT-ACC here and in (27) may be treated as a KP causer that covers a PP causer as in *The window broke from the storm* (Alexiadou et al. 2015: 8-9). Such cases are also subject to Pytkäinen's (2008) Low Applicative analysis, wherein a transfer of possession takes place between ACC and DAT, DAT being a potential possessor, but not a causer. Many open questions remain here.

¹⁴ In the case of transitives, the case of the causee, the single argument of V/R, can only be ACC, whether a "make-causative" meaning or "let-causative" one is produced. In this respect, Mongolian differs from Japanese, both being Altaic (but controversially). In Japanese, the "make-causative" and "let-causative" meanings are reflected by ACC (the particle *-o*) and DAT (the particle *-ni*), respectively.

3.4. The interface and distinction of syntactic vs. lexical causatives

Another important distinction that must be made is one of syntactic and lexical causatives. Several differences are observed as follows.

Syntactic causatives, employing *-uul* and *-lg*, are highly regulative and productive, while lexical causatives, employing *-uul*, *-lg*, *-aa* and *-l*, are not. Syntactic causatives resist the alternating suffixes *-aa* and *-l*, while lexical causatives do not. Janhunen (2012: 149) also observes that the causative suffixes involving a greater degree of lexicalization *-aa* always precede the relatively more productive suffixes *-lg* and *-uul*.

Passivization of syntactic causatives is not allowed but that of lexical causatives is good.

- (42) Bi Badma-g ir-üül-v.
I Badma-ACC come-CS-PST
'I asked Badma to come.'
- (43) *Badma ir-üül-gd-v.
Badma-NOM come-CS-PS-PST
'Int. Badma was asked to come.'
- (44) Bid daisan-ig sön-öö-v.
we enemy-ACC die out-CS-PST
'We eliminated the enemy.'
- (45) Daisan biden-d sön-öö-gd-v.
enemy-NOM by us die out-CS-PS-PST
'The enemy was eliminated by us.'

Syntactic causatives come with scope ambiguity when involving adverbials such as *öörtöö* 'oneself', *zoriud* 'deliberately' and *daxin* 'again', while no scope ambiguity is available in the case of lexical causatives.

- (46) Bi Badma-g zoriud ir-üül-v.
I Badma-ACC deliberately come-CS-PST
'I asked Badma to deliberately come.'
'I deliberately asked Badma to come.'
- (47) Bid daisan-ig zoriud sön-öö-v.
we enemy-ACC deliberately die out-CS-PST
'We deliberately eliminated the enemy.'
'*We made the enemy deliberately die out/be eliminated.'

For syntactic causatives of transitives, transmission of energy is not observed between the causer (e.g. the doctor) and the patient in (49), while for lexical causatives, it is observed between the two arguments (e.g. the nurse and the patient in (48)).¹⁵

¹⁵ See Hashimoto (2019) for detailed discussion.

- (48) *Suvilagč övčtön-ig bos-goo-v.*
 nurse-NOM patient-ACC rise-CS-PST
 ‘The nurse sat the patient up.’
- (49) *Emč suvilagč-aar övčtön-ig bos-goo-lg-v.*
 doctor-NOM nurse-INS patient-ACC rise-CS-CS-PST
 ‘The doctor made the nurse sit the patient up.’

3.5. A formal characterization of causative formation

One of the most important issues concerning formal characterization of causative formation is syntactic categorization and specification of a causer-introducing head. According to the category of the root word, causative formation in Mongolian falls into five patterns.

- (50) a. $V_{in} \rightarrow V_{tr}$
 b. $V_{tr} \rightarrow V_{tr}$
 c. $N \rightarrow V_{in} \rightarrow V_{tr}$
 d. $Adj \rightarrow V_{in} \rightarrow V_{tr}$
 e. $Adv \rightarrow V_{tr}$

In pattern 1, a light verb, which is spelled out as CS, selects for an intransitive verb V_{in} to form a transitive verb V_{tr} . A transitivizing head is assumed to be v (Chomsky 1995) or Voice (Kratzer 1996), which introduces an external argument EA. Since $V_{in} \rightarrow V_{tr}$ yields either a lexical or a syntactic causative, whose formations are of different levels, it is reasonable to assume that causers are introduced by different heads. I then, following Nie (2020), Aoyagi (2021) and Bai (2022), postulate a split VoiceP (vP) consisting of separate projections. The lower projection is labeled “VoiceP1” and the higher one “VoiceP2”. Voice1 is spelled out as a lexical CS and Voice2 as a syntactic CS. The separation of two Voice heads is evidenced by verbs like *bos-goo-lg*, which means “X make Y cause Z to stand/sit up”.

(51) Pattern 1: $V_{in} \rightarrow V_{tr}$

- a. [_{VoiceP2} [_{VoiceP1} Causer [_{VP} IA V] -CS]] (lexical causative)
 b. [_{VoiceP2} Causer [_{VoiceP1} [_{VP} IA V]] -CS] (syntactic causative)

In pattern 2, V merges with the internal argument IA to form an intransitive verb phrase VP, which in turn merges with EA-introducing head Voice1 and the resulting phrase is selected by the causative head Voice2, which introduces the causer. This formation applies to type 3, 4 and 5 in (28). As for type 2, the causer is introduced by Voice1 since type 2 is lexical causatives. The DAT argument is merged lower than Voice1 since it does not contribute to causativization. As the DAT argument is not a core one of V , it behaves also like an adjunct. This allows us to treat it as an applied argument. Following Pyllkannen (2008), I argue that an applied argument lower than EA is introduced by a low applicative head.

(52) Pattern 2: $V_{tr} \rightarrow V_{tr}$

- a. [VoiceP2 Causer [VoiceP1 EA [VP IA V]] -CS] (syntactic)
- b. [VoiceP2 [VoiceP1 Causer [AppP DAT/ACC [VP IA V] -CS]]] (lexical)

In pattern 3, there are two verbal affixes, one spelling out the head that takes care of the inchoative interpretation and the other spelling out the causative head Voice1. IA is selected for by the derived intransitive verb (N plus verbalizing suffix INCH) to form VP, which is in turn selected for by Voice1, spelled out as CS, which introduces the causer.

(53) Pattern 3: $N \rightarrow V_{in} \rightarrow V_{tr}$

- [VoiceP2 [VoiceP1 Causer [VP IA [NP N] -INCH] -CS]] (lexical)

Pattern 4 differs from pattern 3 in that it has the root merged as an adjective. Of course, the adjective can be a derived one. However, in the context of Distributed Morphology, especially of Root syntax, what we label “N”, “Adj”, “Adv” or “V” would all be treated as a categoryless element R. The technical treatment of the root word, however, is not within the scope of this paper.

(54) Pattern 4: $Adj \rightarrow V_{in} \rightarrow V_{tr}$

- [VoiceP2 [VoiceP1 Causer [VP IA [AdjP Adj] -INCH] -CS]] (lexical)

In pattern 5, CS *-l* spells out the causative head Voice1, which selects for VP headed by an inchoative head, roughly the same as BECOME as often assumed in the literature. This head is spelled out as *-r* when not selected for by Voice1. That is, *-l* and *-r* do not co-occur. Yet they are realizations of different heads; they are not competitive. The head realized as *-r* is a verbalizer while the one realized as *-l* is not. Another possibility is that they both are verbalizers. If this is the case, no inchoative head is postulated below CS *-l*, as represented in (55b). The verbalizing function is encoded by the causative head Voice1. That is, Voice1 and the verbalizing head, say, V, are bundled as a complex head Voice1+V, which is spelled out as *-l*. This analysis is more reasonable given that *-l* and *-r* do not co-occur in the same verb. Under this analysis, however, IA is forced to be merged with Adv first to form AdvP, which in turn merges with Voice1+V. This is compatible with the standard assumption that the object is selected by R (Adv here) first, not with RP.¹⁶

(55) Pattern 5: $Adv \rightarrow V_{tr}$

- a. [VoiceP2 [VoiceP1 Causer [VP IA [AdvP Adv] -INCH] -CS]] (lexical)
- b. [VoiceP2 [(Voice1+V)P Causer [AdvP IA Adv] -CS]] (lexical)

¹⁶ If what we take to be Adv is in fact R, a categoryless element, and merges with INCH rather than with IA, then this exemplifies Embick’s (2023: 4) Root Hypothesis (roots do not project phrasal structure).

4. Semantics

In this section, I exemplify the nine parameters of causatives presented in Dixon (2000).

Parameter 1. State/action. Whether a causative mechanism applies only to state verbs or also to action verbs. In Mongolian, it seems that action verbs are more natural as input to causativization. This is suggested by the fact that if a given verb V can either be a state verb or be an action verb, V-cs can only denote causation of an action, as exemplified below. Often, but not always, V-cs has an idiosyncratic meaning or even comes with semantic drift.

(56) Causativization of verbs denoting an action and a state

<i>sana</i>	‘think of’	‘miss’
<i>sana-uul</i>	‘remind (i.e. make think of)’	#‘make miss’
<i>bod</i>	‘caculate’	‘think/consider’
<i>bod-uul</i>	‘cause to caculate’	#‘cause to think/consider’
<i>xairla</i>	‘fall in love’	‘like’
<i>xairla-uul</i>	‘cause to fall in love’	#‘cause to like’

Verbs of existence or with an existential nuance, however, seem to be quite natural as input to causativization.

(57) Causativization of verbs with an existential nuance

<i>bai</i>	‘be’	<i>üld</i>	‘remain’
<i>bai-lg</i>	‘let be’	<i>üld-ee</i>	‘retain’
<i>taar</i>	‘go right’	<i>bairl</i>	‘be located in’
<i>taar-uul</i>	‘get something right’	<i>bairl-uul</i>	‘find a place for’

Parameter 2. Transitivity. Whether a causative mechanism applies only to intransitive verbs, or to both intransitive and simple transitive verbs, or to all types of verbs – intransitive, simple transitive and also ditransitive. As discussed in section 3, the Mongolian causative mechanism applies to both intransitive and simple transitive verbs. It applies also to ditransitive verbs and derived causative verbs, as shown below. In the latter case, *-aa* and *-l* are not available.

- (58) Bi Bat-aar ter nom-ig Badma-d ög-үүл-v. (causativization of ditransitive)
I Bat-INS that book-ACC Badma-DAT give-CS-PST
‘I asked Bat to give the book to Badma.’

Parameter 3. Control. Whether the causee lacks control or has control of the activity. In Mongolian causatives, the activity can be either controlled or not by the causee and this is reflected, in the case of V being a transitive, by differential case marking to some extent. When

the causee is marked by INS, it can have control of the activity and this is not true in the case of ACC. For example, in (59) V-cs denotes an action that is almost completely controlled by the subject while in (60) the embedded subject has control of the activity to some extent. In the former case, the activity, which is under the control of the subject on a large part, already involves the subject besides the object so that *id-üül* means “dine”, rather than “cause to eat”.

- (59) Bi naiz nar-aa id-üül-v.
 I friends-ACC-RX eat-CS-PST
 ‘I dined my friends.’
- (60) Bi Bars-aar con-ig id-üül-v.
 I tiger-INS wolf-ACC eat-CS-PST
 ‘I made the tiger eat the wolf.’

As also discussed by Hou (1992: 471), ACC is assigned when the force of manipulation by the causer is stronger and the force of action of the causee is weaker.¹⁷ DAT seems to be less sensitive to the causee’s controllability of the activity than INS and ACC.

Parameter 4. Volition. Whether the causee performs the action willingly (‘let’) or unwillingly (‘make’). This parameter is arbitrary in Mongolian. Interestingly, when V is an intransitive, the same case ACC is employed whether the causee is willing to perform the action or not, unlike in languages like Japanese, in which differential case marking is employed according to the causee’s (un)willingness. When V is a transitive, INS can be employed in either case. The same is true in the dative causative. It then follows that in Mongolian causatives, the causee’s (un)willingness is not exponent as differential Case.

Parameter 5. Affectedness. Whether the causee is only partially affected or completely affected by the activity. This parameter is dumb in Mongolian.¹⁸

Parameter 6. Directness. Whether the causer acts directly or indirectly. As exemplified below, the causer directly performs the action in the dative causatives and accusative causatives while it indirectly performs the action in the instrumental causative.

- (61) Bi xüüxed-ee bičig üz-üül-v.
 I child-ACC-RX attend school-CS-PST
 ‘I made my children attend school.’
- (62) Ter emč-d övčin-öö üz-üül-v.
 he doctor-DAT illness-ACC-RX see-CS-PST
 ‘He had seen a doctor.’
- (63) Ter emč-eer övčin-öö üz-üül-v.
 he doctor-INS illness-ACC-RX see-CS-PST

¹⁷ For discussion on the causer’s controllability of the action, see Svantesson (2003: 171-172), Janhunen (2012: 249) and Hashimoto (2019: 145; 186, fn.19).

¹⁸ Affectedness is an aspectual notion in a large part and in Mongolian it is partially expressed by the aspectual suffix *-čix*. Note that affectedness discussed by Dixon’s (2000) refers to exhaustedness in some sense and is not the same as the “affectedness” discussed by Hashimoto (2019) on Mongolian causatives.

‘He asked a doctor to treat himself.’ (Lit. ‘He made a doctor see his illness.’)

Parameter 7. Intention. Whether the causer achieves the result accidentally or intentionally. This parameter concerns type 4 and 5 in (28). As discussed earlier, the causer can be adversely affected by the agent’s performing the action on the patient. When adversely affected, the causer achieves the result accidentally and when not adversely affected, it achieves the result intentionally.

Parameter 8. Naturalness. Whether the activity happens fairly naturally (the causer just initiating a natural process) or it is achieved only with effort (perhaps, with violence). In Mongolian, this parameter works for comparison of plain transitives (V) and causatives derived out of their (fake) passives (V-PS-CS). In (64), the activity takes place naturally but it does not in (65), which is not a causative, though. V-PS-CS mostly denotes activities that are natural processes not controlled by the causer.

(64) Causative denoting activity that naturally happens:

<i>nem</i>	‘add’	<i>bod</i>	‘think of’
<i>nem-gd</i>	‘increase (intr)’	<i>bod-gd</i>	‘come to mind’
<i>nem-gd-üül</i>	‘increase (tr)’	<i>bod-gd-uul</i>	‘remind of’

(65) Causative denoting activity that is achieved with effort:

<i>bari</i>	‘arrest’	<i>šat</i>	‘burn (intr)’
<i>bari-uul</i>	‘cause to arrest’	<i>šat-ta</i>	‘burn (tr)’

Parameter 9. Involvement. Whether the causer is also involved in the activity (in addition to the causee) or not involved. This parameter concerns the distinction of lexical and syntactic causatives most. *Bos-goo* denotes an action of standing or sitting someone up in which the causer is also the agent and therefore is involved in the action. By contrast, *bos-goo-lg* denotes an action of asking someone to stand or sit someone else up in which the causer is not the agent and therefore is not involved in the action.

(66) *Suvilagč* *övčtön-ig* *bos-goo-v.*
nurse-NOM patient-ACC rise-CS-PST
‘The nurse sat the patient up.’ (=48)

(67) *Emč* *suvilagč-aar* *övčtön-ig* *bos-goo-lg-v.*
doctor-NOM nurse-INS patient-ACC rise-CS-CS-PST
‘The doctor made the nurse sit the patient up.’ (=49)

Importantly, (un)involvement is not the defining property of lexical and syntactic causatives. In the following example, which is a lexical causative, the causer may be a different person than the very one who performs the action of burning the old books. That is, the causer instantiates and controls the action but not performs it in person. This sentence is perfectly

good in a context in which, for example, I asked John to help me burn the books as a housekeeping service.

- (68) Bi xuučin nom-uud-aa gal-d šat-aa-v.
I old books-ACC-RX fire-DAT burn-CS-PST
'I burned my old books in the fire (myself).'
'I had my old books burnt in the fire (by someone else).'

Interestingly, this parameter sometimes has to do with the choice between causative morphemes. As shown below, the shorter CS *-aa* is employed to form a causative that is as compact as can be a simple transitive verb, while the longer CS *-uul* is employed to form a less compact causative. As indicated by the English translations, causality is more salient in when *-uul* is employed; it may be the case that the causer achieves the result by doing something else, which is not the case of *-aa*.

- (69) Bi Badma-g xür-gee-v.
I Badma-ACC send-CS-PST
'I sent Badma (to the station).'
- (70) Badma-g urdaar xür-үүл-eed, manus end-ees gar-ii.
Badma-ACC first arrive-CS-CVB we here-ABL set off-VOL
'After Badma arrived there first, let us set off from here.'
(Lit. 'Let Badma arrive first and we set off from here.')

5. Conclusion

Presenting an interface analysis of Mongolian causatives from a typological perspective, this paper have made attempts to explicating morphological, syntactic and semantic properties of them. As regards their morphology, a few hitherto understudied aspects were discussed, including fakeness and authenticity of the so-called "causative suffixes" and their interaction with causative-anticausative alternations. In discussing syntactic characteristics of the causatives, it was demonstrated that role-changing and role-retaining are crucial for identifying causative properties and they interconnect with differential case marking and the variation of grammatical functions. The discussions on syntax also concerned the lexical-syntactic division of Mongolian causatives and formal representations of their formation. Semantics of Mongolian causatives was discussed in terms of the various relationships between the causer or causee and the action/event, which was formulated as nine parameters including State/Action, Transitivity, Control, Volition, Affectedness, Directness, Intention, Naturalness, and Involvement. It was shown that eight out of the nine parameters are concerned in Mongolian.

Serving mainly to outline the typological characteristics of Mongolian causatives, this paper, unfortunately, leaves many issues unsolved, and will hopefully be able to call the

attention of researchers at home and abroad to Mongolian causatives and related issues.

References

- Alexiadou, A., E. Anagnostopoulou and F. Schäfer. 2015. *External Arguments in Transitivity Alternations: A Layering Approach*. Oxford: Oxford University Press.
- Aoyagi, H. 2021. On the causative and passive morphology in Japanese and Korean. *Open Linguistics* 7: 87-110.
- Bai, Chigchi. 2022. A comparative study of passive-causative and Mongolian and causative-passive in Japanese (モンゴル語の「受動-使役形」と日本語の「使役-受動形」との比較考察), In Tanaka T. et al. (eds.) *Exploring the Nature of Language from Synchronic and Diachronic Perspectives*, 284-291.
- Bai, Chigchi. 2023. A diachronic change in Mongolian passives viewed from *nemegde*- ('Нэмэгд' гэдэг үгээс монгол хэлний үйлдэгдэх гэвийн түүхэн хувиралтыг ажиглах нь). *Mongolian Language* 540: 27-31.
- Chomsky, N. 1981. *Lectures on Government and Binding*. Dordrecht, The Netherlands: Foris.
- Chomsky, N. 1995. *The Minimalist Program*. Cambridge: MIT Press.
- Cinggeltei. 1999. *Grammar of Modern Mongolian* (Одо үеийн монгол хэлний зүй). Hohhot: Inner Mongolia People's Publishing House.
- Comrie, Bernard. 1976. The syntax of causative constructions: Cross-language similarities and divergences. In M. Shibatani (eds.), *The Grammar of Causative Constructions*, 261-312. London: Academic Press.
- Daobu. 2007. On voice suffixes of Mongolian verbs (蒙古语动词“态”语缀探析). *Minority Languages of China* 5: 3-11.
- Dixon, R.M.W. 2000. A typology of causatives: Form, syntax and meaning. In Dixon, R.M.W., & Alexandra Y. Aikhenvald (eds.), *Changing Valency: Case Studies in Transitivity*, 30-83. Cambridge, MA.: The MIT Press.
- Embick, D. 2023. Smaller syntax for English stative passives: A first report. *Acta Linguistica Academica* 70: 285-316.
- Frawley, W. 1992. *Linguistic semantics*. Hillsdale, N.J.: Lawrence Erlbaum.
- Hashimoto, Kunihito. 2019. The Different Meanings of Causative Sentences in Mongolian (モンゴル語の使役文の意味の違いについて). *Journal of Language and Culture of Hokkaido*, 17: 141-188.
- Haspelmath, M. 1993. More on the typology of inchoative/causative verb alternations. In Comrie, B. & M. Polinsky (eds.) *Causatives and Transitivity*, 87-120. Amsterdam: John Benjamins.
- Hou, Wanzhuang. 1992. On the grammatical and semantic structures of causative sentences in Mongolian (Монгол хэлний бусдаар үйлдүүлэх өгүүлбэрийн хэл зүйн бүтэц ба утга зүйн бүтцийн тухай өгүүлэх нь). *Proceedings of Conference of China's Mongolian Language Society* 2: 451-474.

- Janhunen, J. A. 2012. *Mongolian*. Amsterdam/Philadelphia: John Benjamins.
- Kratzer, A. 1996. Severing the external argument from its verb. In J. Rooryck and L. Zaring (eds.), *Phrase Structure and the Lexicon*. Dordrecht: Kluwer, 109-137.
- Kullman, R. and Tserenpil. 2005. *Mongolian Grammar*, 5th edition, Switzerland: Kullnom Verlag.
- Nasanbayar. 1982. *Modern Mongolian* (Орчин цагийн монгол хэл). Hohhot: Inner Mongolia Education Press.
- Nie, Yining. 2020. Morphological causatives are voice over voice. *Word Structure* 13: 102-126.
- Poppe, N. 1954. *Grammar of Written Mongolian*. Wiesbaden: Otto Harrassowitz.
- Pylkkänen, L. 2008. *Introducing Arguments*. Cambridge, Ma.: MIT Press.
- Sanzheyev, G. D. 1973. *The Modern Mongolian Language*. Moskow: Nauka Publishing House.
- Shibatani, M. 1976. The grammar of causative constructions: A conspectus. In M. Shibatani (eds.), *Syntax and Semantics: The Grammar of Causative Constructions*, 1-39. London: Academic Press.
- Song, J. J. 1996. *Causatives and Causation: A Universal-typological Perspective*. London and New York: Addison Wesley Longman.
- Svantesson, J. 2003. Khalkha. In Janhunen J. (eds.) *The Mongolic Languages*, 154-176. London: Routledge.

Appendix: List of morph- and sem-causative verbs discussed in the paper

Latin	Cyrillic	Uyghur	Latin	Cyrillic	Uyghur
av-uul	ав-уул	اۋۇل	inee-lg	инээ-лгэ	ينەئەلگە
orč-uul	орч-уул	ورچۇل	xii-lg	хий-лгэ	خىيەلگە
unš-uul	унш-уул	ۋنشۇل	težee-lg	тэжээ-лгэ	تەجەئەلگە
bairl-uul	байрл-уул	بايرلۇل	šat-aa	шат-аа	شاتاا
bari-uul	бари-ул	بارىۇل	öld-ee	үлд-ээ	ۆلدەئە
bod-uul	бод-уул	بودۇل	xürt-ee	хүрт-ээ	خۇرتەئە
bodgd-uul	бодогд-уул	بودوگدۇل	sön-öö	сөн-өө	سەنөө
xairla-uul	хайрл-уул	خايرلۇل	aild-ga	айлт-га	ايلتغا
sana-uul	сан-уул	سانۇل	bags-ga	багас-га	باغاسغا
taar-uul	даар-уул	تاارۇل	sur-ga	сур-га	سۇرغا
zagn-uul	загн-уул	زاغنۇل	bos-go	бос-го	بوسگو
zas-uul	зас-уул	زاسۇل	xür-ge	хүр-г	خۇرگە
bai-gul	байг-уул	بايگۇل	ebd-l	эвдэ-л	ەبدەل
erg-үүл	эрг-үүл	ەرگүүл	balb-l	балба-л	بالبال
id-үүл	ид-үүл	يدүүл	bialb-l	бялба-л	بىالبال
ir-үүл	ирүүл	يرүүл	xag-l	хага-л	خاغال

ög-үүл	өг-үүл	ᠣᠭᠦᠭᠦᠯ	xug-l	хуга-л	ᠬᠤᠭᠠᠯ
üz-үүл	үз-үүл	ᠦᠵᠦᠭᠦᠯ	sox-l	сохо-л	ᠰᠣᠬᠤᠯ
xel-үүл	хэл-үүл	ᠬᠡᠯᠦᠭᠦᠯ	nebt-l	нэвтэ-л	ᠨᠡᠪᠲᠡᠯ
xögž-үүл	хөгж-үүл	ᠬᠥᠭᠵᠡᠭᠦᠯ	niac-l	няца-л	ᠨᠢᠴᠠᠯ
xür-үүл	гөр-үүл	ᠬᠦᠷᠦᠭᠦᠯ	tas-l	таса-л	ᠲᠠᠰᠠᠯ
nemgd-үүл	нэмэгд-үүл	ᠨᠡᠮᠡᠭᠳᠡᠭᠦᠯ	tod-l	тодо-л	ᠲᠣᠳᠤᠯ
med-үүл	мэд-үүл	ᠮᠡᠳᠡᠭᠦᠯ	coo-l	цоо-л	ᠴᠣᠣᠯ
zeel-үүл	зээл-үүл	ᠵᠡᠭᠡᠯᠦᠭᠦᠯ	cöm-l	цөмө-л	ᠴᠥᠮᠥᠯ
uu-lg	уу-лга	ᠤᠭᠤᠯᠭᠠ	zad-l	зада-л	ᠵᠠᠳᠠᠯ
bai-lg	бай-лга	ᠪᠠᠶᠤᠯᠭᠠ	bar	бар	ᠪᠠᠷ
bosgo-lg	босго-лга	ᠪᠣᠰᠭᠣᠯᠭᠠ	nee	нээ	ᠨᠡᠭᠡ
cai-lg	цай-лга	ᠴᠠᠶᠤᠯᠭᠠ	xaa	хаа	ᠬᠠᠠ
xania-lg	ханиа-лга	ᠬᠠᠨᠢᠶᠠᠯᠭᠠ			